



Integrating domain knowledge into transformer for short-term wind power forecasting

Junhao Cheng^a, Xing Luo^{b,*}, Zhi Jin^{a,c,*}

^a School of Intelligent Systems Engineering, Shenzhen Campus of Sun Yat-sen University, Shenzhen, Guangdong, 518107, PR China

^b Department of Frontier Research, Peng Cheng Laboratory, Shenzhen, 518055, PR China

^c Guangdong Provincial Key Laboratory of Fire Science and Technology, Guangzhou, 510006, PR China

ARTICLE INFO

Keywords:

Wind power forecasting

Deep learning

Domain knowledge

Domain-knowledge integrated transformer model

ABSTRACT

Wind energy is an environmentally friendly source of energy and serves as an efficient supplement to conventional energy resources. Accurate wind power forecasting is crucial for effective decision-making in the daily operation of wind power plants. However, due to the heavy dependence on weather conditions, the variability and uncertainty associated with weather pose significant challenges to wind power forecasting. In this study, we propose a domain-knowledge integrated Transformer (DKFormer) model for short-term wind power forecasting. The proposed model integrates domain knowledge of wind power generation through three portable modules that play essential roles in data pre-processing, model training, and forecasting stages respectively. Additionally, by constructing boundary constraints that simultaneously utilize the data of both measured wind power and numerical weather prediction (NWP), the DKFormer model further reduces errors in multi-step wind power forecasting and improves overall forecast performance, particularly when input wind speed data exhibits dramatic variations. Furthermore, transfer learning techniques are employed to enhance the forecast capability of the DKFormer model using limited training data. Real-life datasets are used to evaluate the performance of the proposed DKFormer, demonstrating its superiority over conventional statistical models and DL models in short-term wind forecasting. Specifically, in day-ahead wind power forecasting experiments, our proposed DKFormer model achieves a 22.0% reduction in mean absolute error (MAE) while also exhibiting improved forecast stability compared to the conventional Transformer model.

1. Introduction

Wind power is widely acknowledged as an environmentally friendly and clean energy source, distinguishing itself from fossil fuels by its negligible contribution to greenhouse gas emissions and air pollutants that may exacerbate climate change. The remarkable global expansion of wind power utilization serves as a testament to its prominent status as one of the most promising sources of clean energy. Between 2012 and 2022, total global wind power generation experienced an average growth rate of 14.8%, reaching 21,048 billion kilowatt-hours by the end of 2022 [1]. However, the inherent characteristics of wind power generation such as stochasticity, intermittency, and uncertainty pose challenges to the secure and stable operation of wind clusters [2]. Consequently, the implementation of forecasting systems within wind clusters assumes critical importance in enabling the formulation of appropriate strategies for power generation, distribution, and maintenance [3,4].

It is widely recognized that feature selection and forecasting models play crucial roles in wind power forecasting. Regarding feature selection, the prevailing methods primarily rely on two types of data: historically measured wind power data and numerical weather prediction (NWP) data. NWP, provided by meteorological departments, offers forecasts of wind speed and other meteorological factors up to 48 h in advance. Leveraging this lead time, NWP-based approaches generate wind power forecasts by utilizing predicted wind speed and the wind power curve [5,6]. However, due to high computational costs, complexity, and concerns about the accuracy of NWP, historical wind power data is more commonly utilized for short-term (within 48 h) wind power forecasting [7–10].

Existing wind power forecasting models can be classified into three primary categories: physical models, statistical models, and deep learning (DL) models. Physical models utilize the Navier–Stokes equation and principles of fluid mechanics to construct a wind field simulation

* Corresponding authors.

E-mail addresses: luox@pcl.ac.cn (X. Luo), jinz26@mail.sysu.edu.cn (Z. Jin).

<https://doi.org/10.1016/j.energy.2024.133511>

Received 5 April 2024; Received in revised form 12 August 2024; Accepted 17 October 2024

Available online 28 October 2024

0360-5442/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

model, enabling the estimation of wind speed and power generation [11]. However, these models often necessitate substantial computational resources and intricate parameter configurations for modeling each individual wind cluster. Additionally, their forecasting performance tends to deteriorate due to complex terrain and unfavorable meteorological conditions [12].

Moreover, statistical models primarily rely on historical wind speed data for analyzing wind power generation [13]. These models are relatively straightforward, adaptable to diverse meteorological conditions, and capable of capturing specific data patterns and trends [14]. Commonly used statistical models in wind power forecasting encompass the auto-regressive moving average (ARMA) model [15], the auto-regressive integrated moving average (ARIMA) model [16], the support vector machine (SVM) [17], the K-nearest neighbors (KNN) model [18], the Kalman filters model [19], the Gaussian process model [20], and the fuzzy logic model [21]. However, statistical models often depend on stringent mathematical relationships between input and output variables which may restrict their ability to handle complex nonlinear associations between wind power and various input variables as well as sudden events caused by abrupt changes in wind speed. Consequently, there is a general decline in forecasting accuracy with an increasing time span [22].

Furthermore, with the rapid advancement of artificial intelligence technology, an increasing number of researchers are turning to DL models for wind power forecasting. By harnessing high computational power and employing more intricate network structures, these models can often achieve remarkable accuracy even without considering specific domain knowledge. Among them, recurrent neural networks (RNNs) stand out as exceptional representations due to their ability to handle sequential data, automatically extract features and patterns, and make predictions [23]. Several variants of RNNs have been extensively utilized in wind power forecasting, including long short-term memory (LSTM) [24], bidirectional LSTM (BiLSTM) [25], and gated recurrent unit (GRU) [26]. In 2017, Google researchers introduced a sequence-to-sequence model known as the Transformer which has demonstrated outstanding performance in time-series data-based forecasting tasks [27]. For instance, Qu et al. were pioneers in applying the Transformer model to wind power forecasting by utilizing a linear layer to map wind power time series into an N-dimensional space while removing the SoftMax layer from Transformer [28]. Wang et al. proposed a unified framework called M2TNet that integrates multiple feature extractors such as a Transformer, a feature fusion layer-based fully-connected network, and a prediction terminal layer [29]. Sun et al. on the other hand, employed transformer neural networks to learn sequential dependencies irrespective of distance [30]. The multi-head self-attention mechanism serves as the core component of the Transformer enables multiple attention focuses across different positions thereby allowing the model to effectively capture dependencies over longer distances compared with other DL models.

Although significant progress has been made in artificial intelligence-based wind power forecasting, several challenges persist. First, of utmost importance is the fact that most existing DL models primarily rely on data-driven approaches, with limited incorporation of domain knowledge specific to wind power generation. However, given the intricate physical processes and engineering principles involved in wind power generation, encompassing factors such as wind speed, wind direction, and operational characteristics of turbine blades, the inherent black-box nature of DL methods coupled with a lack of integration of domain expertise may potentially yield predictions inconsistent with fundamental physical principles [23–27,31–34].

Second, there is untapped potential for improvement in harnessing numerical weather prediction (NWP) data and historical wind power information [22,24,28–30]. Existing DL models used for time series forecasting primarily rely on sequential extrapolation techniques to predict future wind power based solely on input features of equal lengths. However, the limitations imposed by their model architectures

restrict methods like LSTM and Transformer to consider only historical wind power or NWP as inputs. Nevertheless, incorporating NWP data into DL models for sequential extrapolation enables them to learn from past patterns and trends while leveraging additional insights provided by NWP data. Another crucial aspect is the presence of significant noise in NWP data; hence finding a suitable denoising algorithm becomes essential. Wang et al. [35] applied wavelet transform (WT) to the original wind power sequence, resulting in an approximation series and three detailed series through nine vanishing moments and three decomposition levels. However, traditional WT has limitations when it come to handling input signal sizes [36]. Abedinia et al. [37] utilized variational mode decomposition (VMD) method and demonstrated that VMD is prone to mode mixing problems in wind power forecasting tasks. Some studies [36,38] have successfully employed empirical mode decomposition (EMD) for denoising; however, challenges related to endpoint effect and mode mixing still persist.

Third, DL models typically require a substantial amount of data to capture the underlying relationships between inputs and outputs [23–27]. If training data is insufficient, these models may not perform as well as some traditional methods [39]. While existing DL models primarily rely on modeling and predicting based on individual wind farm data, there is still significant potential for improving forecasting performance by leveraging adjacent wind farm data sources that have not been fully utilized in previous work.

Considering the aforementioned challenges and drawbacks, we propose a DL model, namely domain-knowledge integrated Transformer (DKFormer), for short-term wind power forecasting. The incorporation of domain knowledge has been taken into account during the development of DKFormer. Furthermore, historical wind power and NWP data are jointly leveraged to guide the forecasting outcomes of DKFormer. Additionally, transfer learning technique is employed to tackle the issue of model training and further enhance the accuracy and stability of DKFormer. The primary contributions of this research can be summarized as follows.

- (1) We initially propose the DKFormer forecasting model, which integrates domain knowledge through three constraint modules that are crucial in data pre-processing, model training, and forecasting stages.
- (2) Unlike conventional sequential extrapolation models that solely rely on historical data for forecasting, our proposed DKFormer model incorporates NWP data's lead time to guide the forecasting results by constructing a cubic polynomial boundary constraint.
- (3) We combine transfer learning technique with the DKFormer to enhance the forecasting performance of our model.
- (4) The proposed domain knowledge constraint modules demonstrate great portability and can be incorporated into various DL baseline models such as LSTM and GRU, resulting in improved forecasting accuracy.

The subsequent sections of this paper are organized as follows: Section 2 introduces the baseline model Transformer, followed by the presentation of our proposed DKFormer model in Section 3. In Section 4, we evaluate the performance of DKFormer using a real-life wind farm dataset and conduct comparative and ablation studies. Finally, the paper is concluded in Section 5.

2. Fundamental transformer model

Since its introduction in 2017, the Transformer model has gained increasing popularity among researchers for various machine learning tasks, including natural language processing (NLP) [27] and computer vision (CV) [40]. In addition, due to its remarkable capability of capturing long-term feature dependencies, the Transformer model has also demonstrated significant value in time series processing [41]. The architecture of the fundamental Transformer model is illustrated in

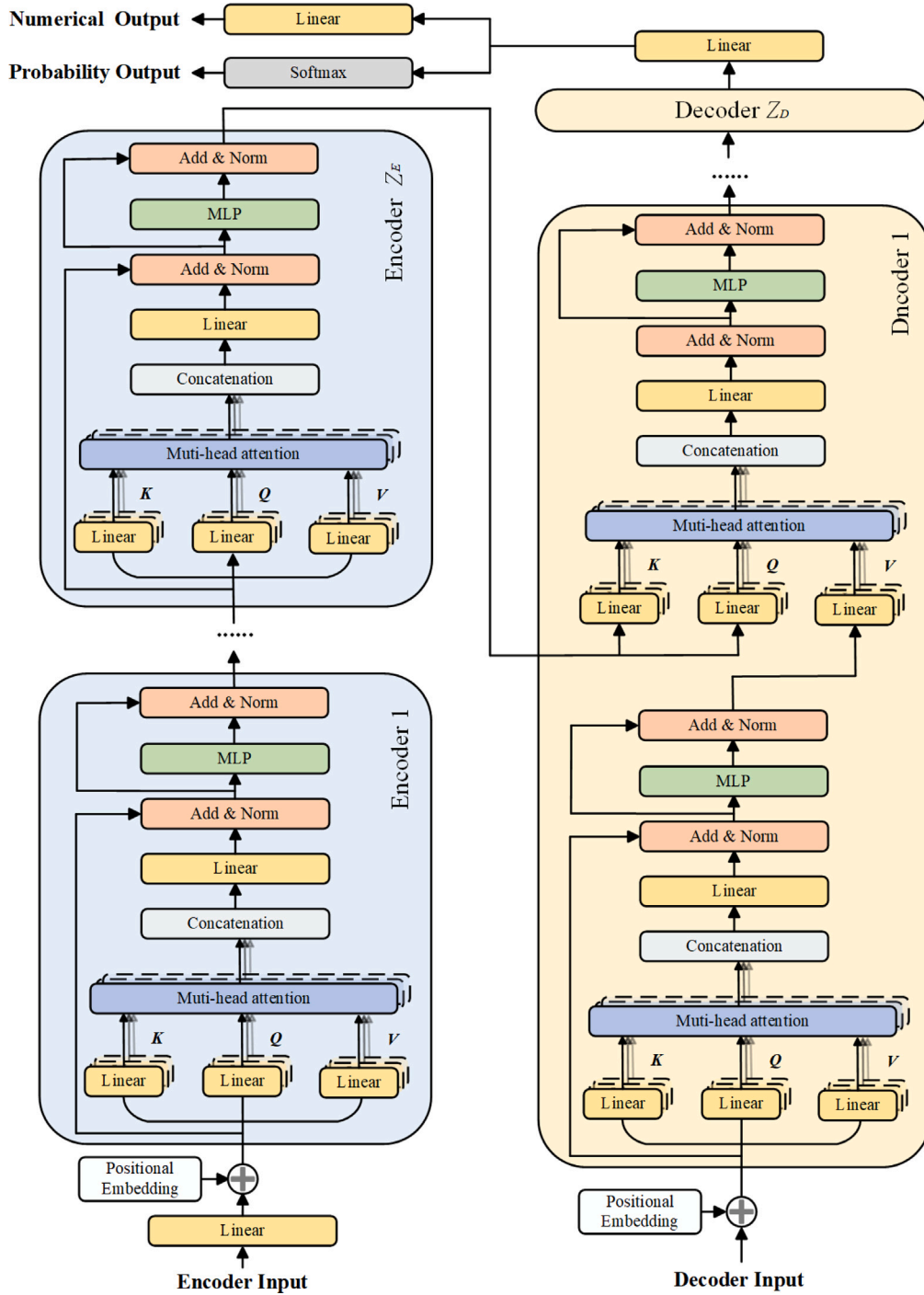


Fig. 1. The architectural design of the Transformer model.

Fig. 1. As described, the encoder and decoder are two indispensable components within the Transformer model.

Specifically, the encoder is responsible for extracting high-level features from the input sequence through multiple encoder modules that encompass a multi-head attention mechanism and a position-wise feed-forward neural network (FFN). The multi-head attention mechanism enables simultaneous attention to diverse positions in the input sequence, while the position-wise FFN conducts non-linear transformations on each position's features. By stacking these encoder modules, the model progressively captures more abstract and global features. On the other hand, the decoder is utilized for generating final predictions and also consists of multiple decoder modules, each comprising two

multi-head attention mechanisms and a position-wise FFN. The decoder dynamically attends to the input sequence by performing self-attention on the encoder outputs and combines them with feature representations from the encoder for forecasting purposes. The multi-head attention mechanisms capture relationships between different positions in the input sequence, while further transforming and adjusting features is achieved through position-wise FFN.

The attention mechanism of the Transformer is its most valuable component, as it effectively captures essential information pertaining to the current task from a vast input corpus [28]. The self-attention mechanism computes attention weights within a sequence and thoroughly examines the structure and interconnections present in the sequence.

Specifically, the attention calculation in the Transformer model incorporates scaling factors and a multi-head mechanism to comprehensively capture crucial latent information from the input data in a multifaceted manner. Mathematically, the scaled dot-product attention employed by Transformer can be represented as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{KQ^T}{\sqrt{d_K}}\right)V \quad (1)$$

In Eq. (1), the matrices Q and K represent queries and keys, respectively, both having a dimension of d_K . The matrices V represent values with a dimension of d_V . The dimensions d_K and d_V are determined by the following equation:

$$d_V = d_K = d_{\text{model}} / h \quad (2)$$

In Eq. (2), d_{model} represents the output dimension of the input layer. The Multi-Head attention mechanism, commonly employed in Transformers, applies h distinct linear transformations to project the input matrices Q , K , and V . Subsequently, the resulting attention outputs are concatenated together. The definition of multi-head attention is as follows.

$$\begin{aligned} \text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \\ \text{head}_i &= \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \end{aligned} \quad (3)$$

where the projection parameter matrices QW_i^Q , KW_i^K , and VW_i^V are denoted as the projection parameters, while h represents the number of attention heads in the attention mechanism.

In contrast to the Transformer models employed in NLP applications, wind power forecasting involves numerical data as the target for prediction. In this scenario, the embedding layer is typically utilized to map the given integers onto an N -dimensional space. Therefore, within DKFormer, a linear layer is employed to perform this mapping for wind power time series, akin to word vectorization in NLP. Furthermore, since the desired output comprises specific numerical values rather than probabilities, the SoftMax layer is omitted from the Transformer and replaced with a Linear layer.

3. Proposed DKFormer model

Section 3 commences by introducing the domain-knowledge integrated Transformer (DKFormer) model and provides a sequential exposition of the three constraint modules in DKFormer, namely the denoising module, clipping module, and boundary module. Subsequently, the transfer learning technique adopted in DKFormer is discussed.

3.1. DKFormer

The architecture of the DKFormer model is depicted in Fig. 2. Prior to model training, the raw wind power data undergoes pre-processing using the denoising module. Throughout the training process, constraints are enforced by means of the clipping module, while leveraging NWP data, the boundary module refines and enhances prediction accuracy. The extraction of training samples, forecasting samples, and their corresponding constraints in DKFormer is illustrated in Fig. 3.

3.1.1. Denoising module

The denoising module, recognized as one of the domain-specific knowledge constraints, is employed to tackle the issue of noise reduction during the data pre-processing stage. Numerical weather prediction wind speed signals often exhibit substantial noise and irregular fluctuations due to factors such as variations in wind direction, terrain obstruction, and sensor errors. These disturbances can significantly undermine the accuracy of forecasting models. To mitigate the impact of noise, we employ ensemble empirical mode decomposition

(EEMD) [42], which is a suitable technique for analyzing and processing non-stationary and non-linear signals, to denoise and extract features from raw wind speed signals. This approach involves conducting multiple empirical mode decomposition (EMD) by superimposing Gaussian white noise onto the signal. By leveraging the statistical properties of Gaussian white noise with uniformly distributed frequencies, different iterations introduce white noise with equal amplitude that modifies extreme points in the signal. Subsequently, averaging intrinsic mode functions (IMFs) obtained from multiple iterations effectively cancels out the added white noise. Introducing randomness in each iteration helps disrupt the patterns present in the original signal, rendering each IMF more independent and distinct, thereby addressing the issue of modal aliasing in EMD. The steps involved in EEMD decomposition are as follows:

1. Add a white noise $n_i(t)$, with a standard normal distribution, to the original signal $x(t)$ to generate a new signal:

$$x_i(t) = x(t) + n_i(t) \quad (4)$$

where, $n_i(t)$ represents the additive white noise sequence for the i th iteration, $x_i(t)$ represents the additional noise signal for the i th trial, where i ranges from 1 to the number of overall averaging iterations M .

2. Perform EMD on the obtained noisy signal $x_i(t)$ separately to obtain their respective IMFs and residuals:

$$x_i(t) = \sum_{j=1}^J c_{i,j}(t) + r_{i,j}(t) \quad (5)$$

where $c_{i,j}(t)$ represents the j th Intrinsic Mode Function (IMF) obtained from the decomposition after the i th addition of white noise, and $r_{i,j}(t)$ is the residual function representing the average trend of the signal. J refers to the number of IMFs.

3. Repeat STEP 2 and 3 N times, decomposing the signal with white noise signals of different amplitudes each time. The resulting collection of IMFs is $c_{1,j}(t), c_{2,j}(t), \dots, c_{N,j}(t)$ where $j = 1, 2, \dots, J$.
4. Calculate the normalized correlation coefficient between the intrinsic mode functions and the original signal:

$$r_j = \frac{\sum_{t=1}^N (y(t) - \bar{y})(c_j(t) - \bar{c}_j)}{\sqrt{\sum_{t=1}^N (y(t) - \bar{y})^2 \cdot \sum_{t=1}^N (c_j(t) - \bar{c}_j)^2}} \quad (6)$$

where r_j represents the correlation coefficient between the j th IMF and the original signal $y(t)$, $t = 1, 2, \dots, N$ represents the sampling points of the signal. Obtain the threshold TH from the standard deviation of the correlation coefficients, TH can be denoted as:

$$TH = \sqrt{\frac{1}{n-1} \sum_{j=1}^n (r_j - \bar{r})^2} \quad (7)$$

5. Filter out k effective IMFs using the correlation coefficient and TH . If $r_j > TH$, keep the j th IMF, and then superimpose them to obtain the denoised signal y' .

$$y'(t) = \sum_{j=1}^k c_j(t) \quad (8)$$

The EEMD decomposition is applied to the NWP data obtained from the wind cluster, with the iteration count and symmetric extension count set at 150 and 4 respectively, following previous research [42]. Fig. 4 illustrates the normalized NWP signal of length 1500 samples alongside the denoised signal obtained after the decomposition. In practical scenarios, when training the DKFormer model, we can access to both its historical and future values. Therefore, applying the denoising module directly to the entire NWP series does not result in data leakage as long as explicit splitting into a training set and a test set is done.

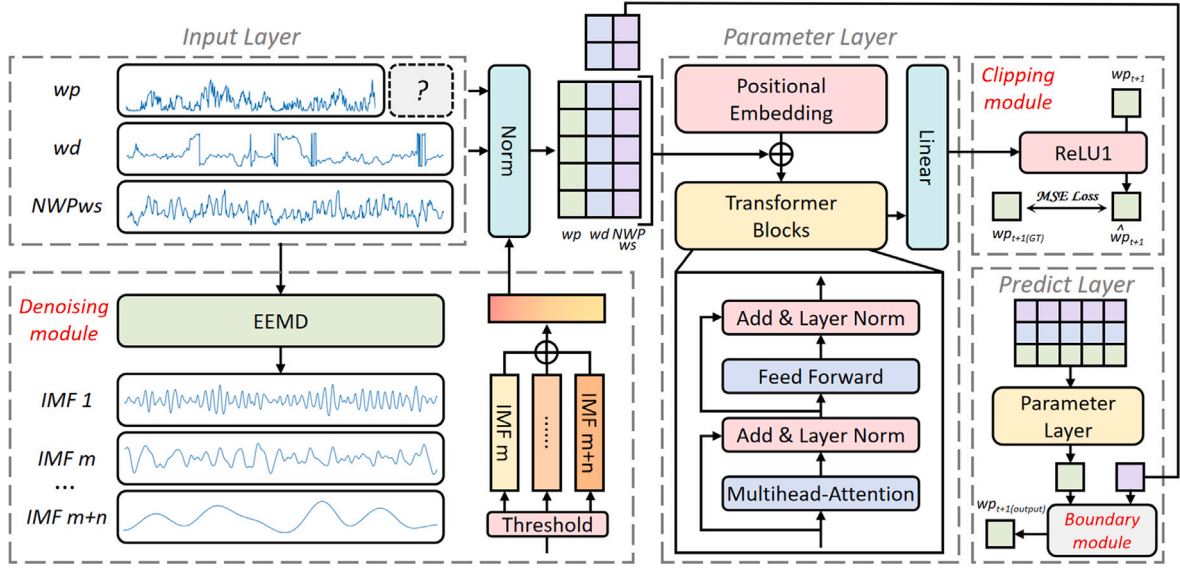


Fig. 2. The architectural design of the DKFormer model for wind power forecasting.

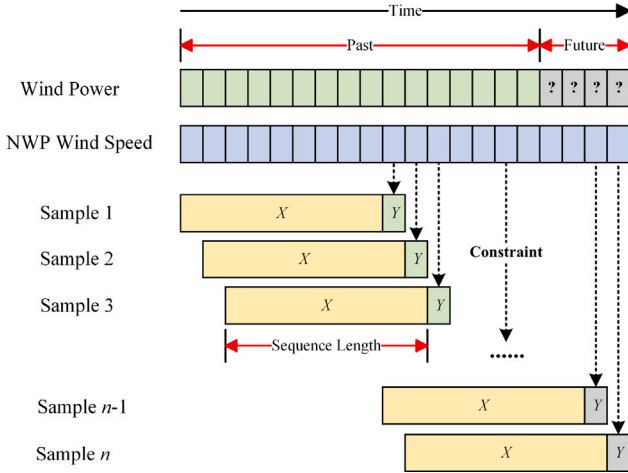


Fig. 3. Schematic representation of training samples, forecasting samples, and their corresponding constraints in DKFormer.

3.1.2. Clipping module

The clipping module integrated into DKFormer plays a crucial role during model training by introducing a new activation function ReLU1 for normalized wind power forecasting, which is inspired by the ReLU function and clips the output to a reasonable range before computing the loss. Based on domain knowledge of wind power generation, it is well-established that the power generation of a wind cluster at any given moment should be a non-negative value within the maximum power limit [43]. Therefore, the predicted output should be constrained by the following equation:

$$\hat{y}_i = \text{ReLU1}(NN(x_i; \theta)) \quad (9)$$

$$\text{ReLU1}(y_i) = \max(0, \min(1, y_i))$$

where x_i stand for the input of DKFormer and y_i is the original output from the neural network. As depicted in Fig. 2, following the acquisition of preliminary predicted values during the model training process, these values are subsequently subjected to the clipping module. This module applies Eq. (9) to ensure their adherence to the constraints, specifically being greater than or equal to 0 and less than the maximum observed power of the wind farm. Subsequently, by comparing these updated values with ground truth values, the loss is calculated

and model parameters are updated through back-propagation. This approach enables DKFormer to effectively acquire domain knowledge and generate predictions that align with the physical laws.

3.1.3. Boundary module

The wind turbines in wind power plants harness the kinetic energy of the air passing through their effective blade area and convert it into electrical energy. The relationship between the output power of a wind turbine and the kinetic energy of the airflow can be described as follows:

$$P_m = \frac{1}{2} C_P \rho A V^3 \quad (10)$$

where P_m is the wind power produced by the wind turbine; C_P is the turbine power coefficient, ρ is the air density, A is the area swept by the turbine blades, and V represents the measured wind speed at the current moment. Due to the uncertainty of NWP, there is an error ϵ in the NWP wind speed. The constraint equation will be transformed into the following form [44]:

$$P_m = \frac{1}{2} C_P \rho A (V_{\text{NWP}} + \epsilon)^3 \quad (11)$$

where V_{NWP} represents the numerical weather predicted wind speed (NWPws). In Fig. 5, the relationship between NWPws and wind power (wp) in actual wind clusters is depicted. It can be observed that the graph of NWPws-wp exhibits distinct boundaries. However, when there is a sudden change in wind speed, the Transformer-based forecasting model often experiences some latency in its predictions. To effectively convey the occurrence of sudden changes to the model, especially for multi-step predictions, constraints are imposed at these boundaries through the boundary module.

The upper and lower limit functions for constraining wind power output predictions during forecasting can be determined using historical data's boundary points (excluding outliers with the Density-Based Spatial Clustering of Applications with Noise algorithm). Since there exists a cubic polynomial relationship between NWPws and wp, cubic polynomials are utilized by the boundary module to fit both upper ($f_{\text{upper}}(x)$) and lower ($f_{\text{lower}}(x)$) boundaries. The analytic forms of bound functions adopted in this work and the fitting results are illustrated specifically in Appendix.

Utilizing the obtained boundaries, DKFormer integrates the single-step forecasting result Y_i into the input of the subsequent step's forecasting X_{i+1} , while also considering the constraint C_{i+1} imposed by NWP

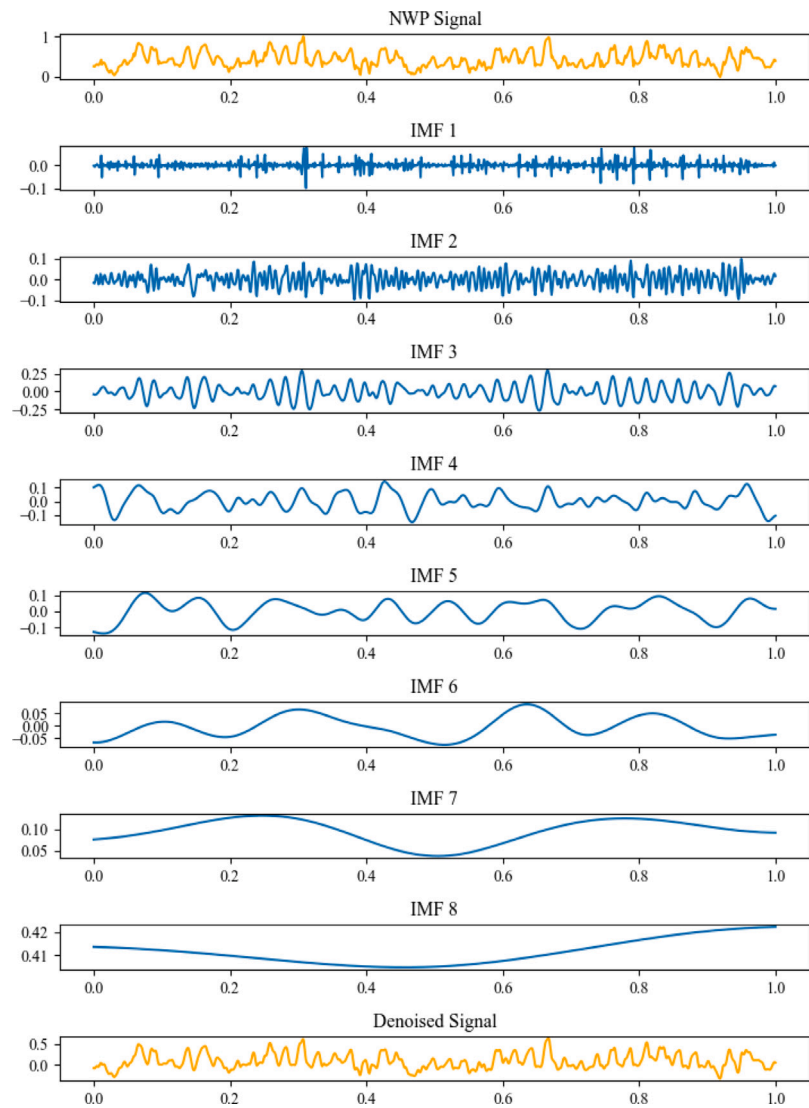


Fig. 4. The schematic diagram of denoising NWP data with EEMD.

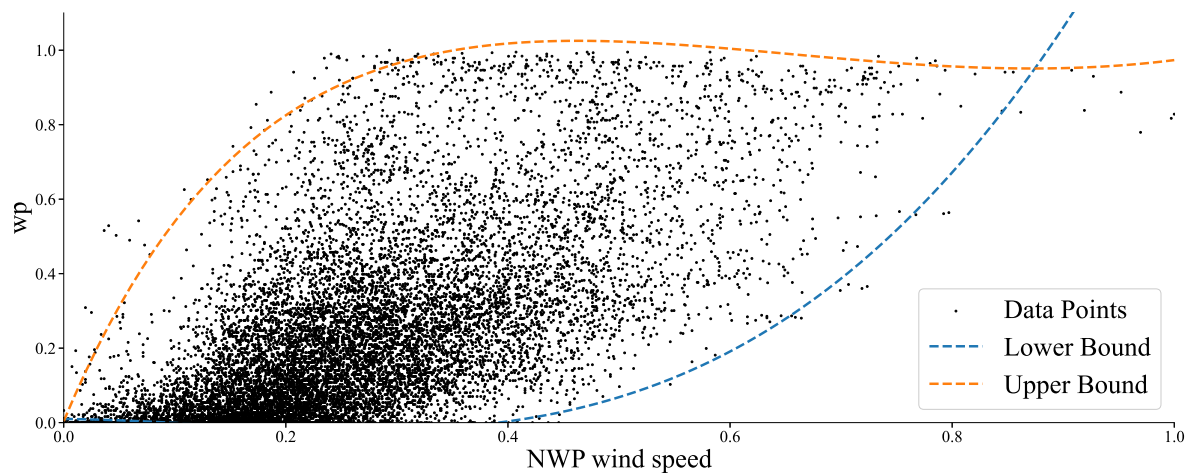


Fig. 5. The relationship and bounds between NWP wind speed and wp on target dataset.

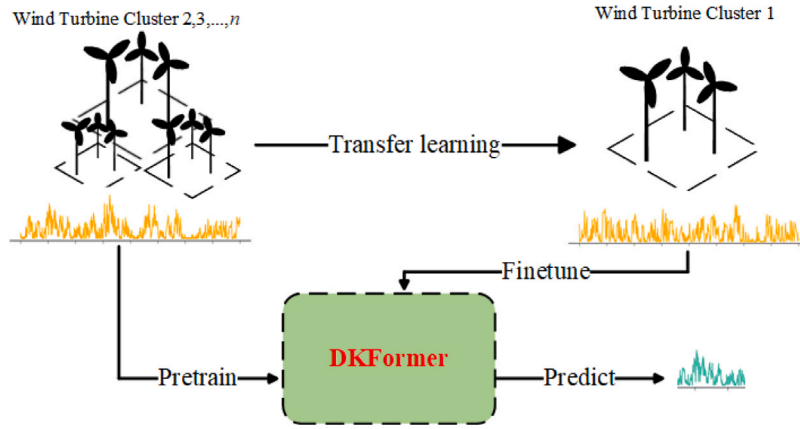


Fig. 6. Schematic diagram of transfer learning.

wind speed. The boundary module rectifies the predicted wind power Y_i through the following formula:

$$Y_{i+1} = \begin{cases} f_{\text{upper}}(C_{i+1}) & \text{if } Y_{i+1} > f_{\text{upper}}(C_{i+1}) \\ Y_{i+1} & \text{if } f_{\text{lower}} \leq Y_{i+1} \leq f_{\text{upper}} \\ f_{\text{lower}}(C_{i+1}) & \text{if } Y_{i+1} < f_{\text{lower}}(C_{i+1}). \end{cases} \quad (12)$$

In this process, any predicted result Y_{i+1} that exceeds the boundaries will be adjusted to the nearest boundary point in order to improve the model's ability to track multi-step predictions using future NWP data.

3.2. Transfer learning

Transfer learning is a pivotal machine learning technique that addresses the inherent challenge of limited training data. The pre-training phase facilitates the extraction of intrinsic patterns within wind power sequences from extensive data collected across multiple wind clusters. Subsequently, the fine-tuning phase focuses on capturing the specific characteristics of an individual wind cluster by utilizing its unique dataset. Leveraging data from multiple wind farms for transfer learning offers significant advantages in terms of enhancing both accuracy and stability in model predictions.

Specifically, in DKFormer, the data from a particular wind cluster is considered as the target domain D_T , while the data provided by other wind clusters serves as the source domain D_S . It should be noted that both the source task T_S and target task T_T are identical.

The process of transfer learning on different wind cluster datasets is depicted in Fig. 6. Initially, DKFormer employs a large-scale dataset from the source domain for model pre-training. Subsequently, the global model is fine-tuned using data from a specific wind farm. To ensure parameter and size alignment, both the pre-training and fine-tuning processes of the global model adopt identical architecture and hyper-parameter selection as discussed in Section 4.2.1 to mitigate overfitting or underfitting issues. Finally, the resulting model is utilized for predicting wind power generation in the target wind farm for the next 1-48 h with an aim to enable capturing similarities in data distribution among various wind farms while also capturing unique patterns specific to each target wind farm. Our findings demonstrate that this transfer learning setup offers advantages in terms of prediction accuracy and stability compared to utilizing data solely from a single wind farm only.

4. Experiment

The proposed DKFormer model is evaluated in this section using real-life datasets provided by [45]. Detailed experiments and analysis are conducted.

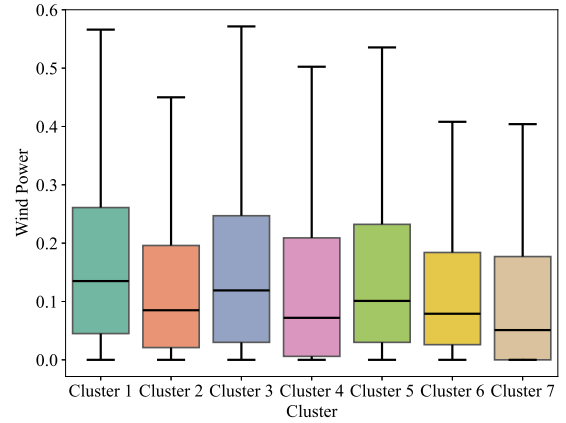


Fig. 7. The distribution pattern of data among the seven wind clusters.

4.1. Data pre-processing

4.1.1. Data description

The adopted datasets comprise seven wind clusters, each characterized by its own unique data distribution, as illustrated in Fig. 7. The historical power measurements are available at an hourly resolution, and the wind power data demonstrates high availability across all wind clusters during the specified period. To ensure confidentiality of commercial secrets, the wind power data is normalized by dividing it by the nominal capacity of each respective wind cluster. Additionally, the provided NWP data includes various meteorological variables, such as longitudinal wind speed (u), radial wind speed (v), total wind speed (ws), and wind direction angle (wd). These weather variables are available from July 1, 2009, to December 31, 2012, with an hourly resolution. For performance evaluation of different models, one wind cluster is randomly selected as the target dataset in accordance with experimental requirements. Following previous work [22], this dataset is partitioned into training, validation, and testing sets in a ratio of 6:1:3 respectively to facilitate the model development and evaluation. Each model undergoes training and validation using distinct data distributions for ten iterations, while being evaluated on a consistent test set.

4.1.2. Data normalization

To address variations in data scales and expedite the convergence speed of the model, a normalization technique is employed. In this study, min-max normalization is chosen due to its extensive utilization in previous research [13,22,30]. The data, including wind power and

u	1	0.019	-0.67	0.41	0.41	0.28
v	0.019	1	-0.19	0.009	0.009	0.14
wd	-0.67	-0.19	1	-0.21	-0.21	-0.14
ws	0.41	0.009	-0.21	1	0.96	0.78
ws'	0.41	0.009	-0.14	0.96	1	0.82
wp	0.28	0.14	-0.14	0.78	0.82	1
	u	v	wd	ws	ws'	wp

Fig. 8. Pearson correlation matrix between different variables.

wind speed, is normalized within the range of [0, 1]. This normalization approach offers several advantages such as facilitating the application of the proposed clipping module that effectively eliminates unreasonable predictions. Additionally, it ensures adherence to established constraints, where wind power and wind speed are always greater than 0 and less than their theoretical maximum value.

The normalization is performed using the following formula:

$$x_{\text{norm}} = \frac{x_{\text{orig}} - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}} \quad (13)$$

where x_{orig} and x_{norm} stand for the original data and the normalized data, respectively. x_{min} and x_{max} represent the maximum and minimum values corresponding to each feature, respectively. It is worth mentioning that the provided wp data has already undergone normalization, thus rendering this step unnecessary for the purposes of this study.

4.1.3. Correlation analysis

It is crucial to evaluate the associations between different variables by utilizing the Pearson correlation coefficient (PCC). The PCC matrix, a widely utilized statistical approach, serves as an effective tool for assessing the linear relationship among variables. Represented in a symmetrical matrix format, each element represents the PCC denoting the association between two variables. Ranging from -1 to 1, this coefficient indicates both the strength and direction of the linear correlation. In this study, data from seven wind clusters are integrated to compute the PCC matrix for the entire dataset, as depicted in Fig. 8. Typically, a PCC equal to or exceeding 0.7 is considered indicative of a strong linear correlation.

According to the results, Fig. 8 illustrates a correlation coefficient of 0.78 between wind speed (ws) and wind power (wp), surpassing the threshold of 0.7. Hence, disregarding the future NWPws data due to their significant association with future wp would be imprudent. This robust correlation between ws and wp further substantiates the rationality and effectiveness of DKFormer' boundary module. Upon applying the denoising module to NWP wind speed data, the correlation coefficient between denoised wind speed (ws') and wp increased to 0.82, providing additional evidence that the EEMD based denoising module can effectively capture the pertinent information in wind speed closely linked to wind power generation. Although other variables exhibit weak correlations with wp, they will also be included as inputs for the DKFormer model.

Table 1

The results of hyper-parameter selection.

α	L	N_{encoder}	N_{decoder}	d_{mlp}	r
0.0001	48	3	4	512	0.15

Fig. 9 depicts the NWPws-wp distributions for all seven wind clusters, while a detailed experiment is specifically conducted on the target wind cluster.

4.1.4. Evaluation metrics

The performance of wind power forecasting models is evaluated in this experiment using commonly employed metrics. The mean absolute error (MAE) and root mean square error (RMSE) are measured in kilowatt (kW), and the R-squared (R^2) is dimensionless. The formulas for these metrics are provided as follows:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (14)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (\bar{y} - y_i)^2}$$

where $\hat{y}_i$ represents the ground truth value, y_i represents the output value, and $\bar{y}$ represents the mean output value.

Standard deviation is commonly employed as a metric to assess the reliability and consistency of model predictions. A smaller standard deviation indicates a higher level of data point concentration, thereby implying greater stability in the model's predictions. The formula of calculating standard deviation is illustrated as follows:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}} \quad (15)$$

where N is the number of samples in the dataset, x_i is the value of the i th sample, and μ is the mean of the samples.

4.2. Experiment setup

4.2.1. Hyper-parameter selection

The proposed model is trained using the Adam optimizer and the mean squared error (MSE) loss function in this experiment. It should be emphasized that the meticulous selection of hyper-parameters significantly influences the model's performance and different choices can lead to substantial variations in its effectiveness. To enhance the efficiency of hyper-parameter selection, a greedy line search strategy is employed, building upon a reference baseline configuration described by [30]. In this approach, each hyper-parameter is individually adjusted within predefined ranges while keeping the other parameters constant. The values of hyper-parameters yielding the lowest prediction errors are separately selected. Variation curves are analyzed to identify optimal performance-inducing hyper-parameter settings.

The DKFormer model necessitates determination of six pivotal adjustable hyper-parameters: learning rate (α), sequence length (L), number of transformer encoders (N_{encoder}) and decoders (N_{decoder}), dimension of MLP layer (d_{mlp}), and dropout rate (r). The outcomes obtained from our greedy experiment are presented in Fig. 10 while the determined values of these hyper-parameters are presented in Table 1. To ensure precise future wind power forecasting, the sequence length (L) is adjusted to 48, enabling the model to assimilate wind power data from the preceding 48 h.

When training DKFormer with these parameters, each epoch takes approximately 34 s, which is comparable to the training time of the basic Transformer model. In addition, DKFormer achieves a prediction speed of 30 ms per step. Although the inclusion of the boundary module

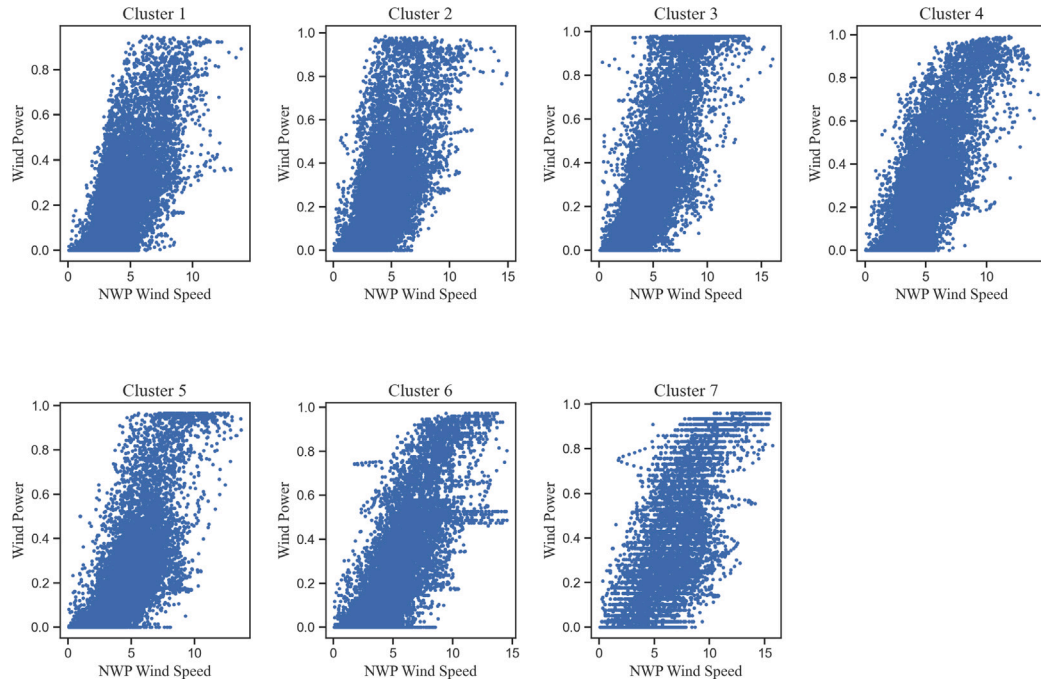


Fig. 9. Data plots between wind speed of NWP and wind power for the seven wind clusters.

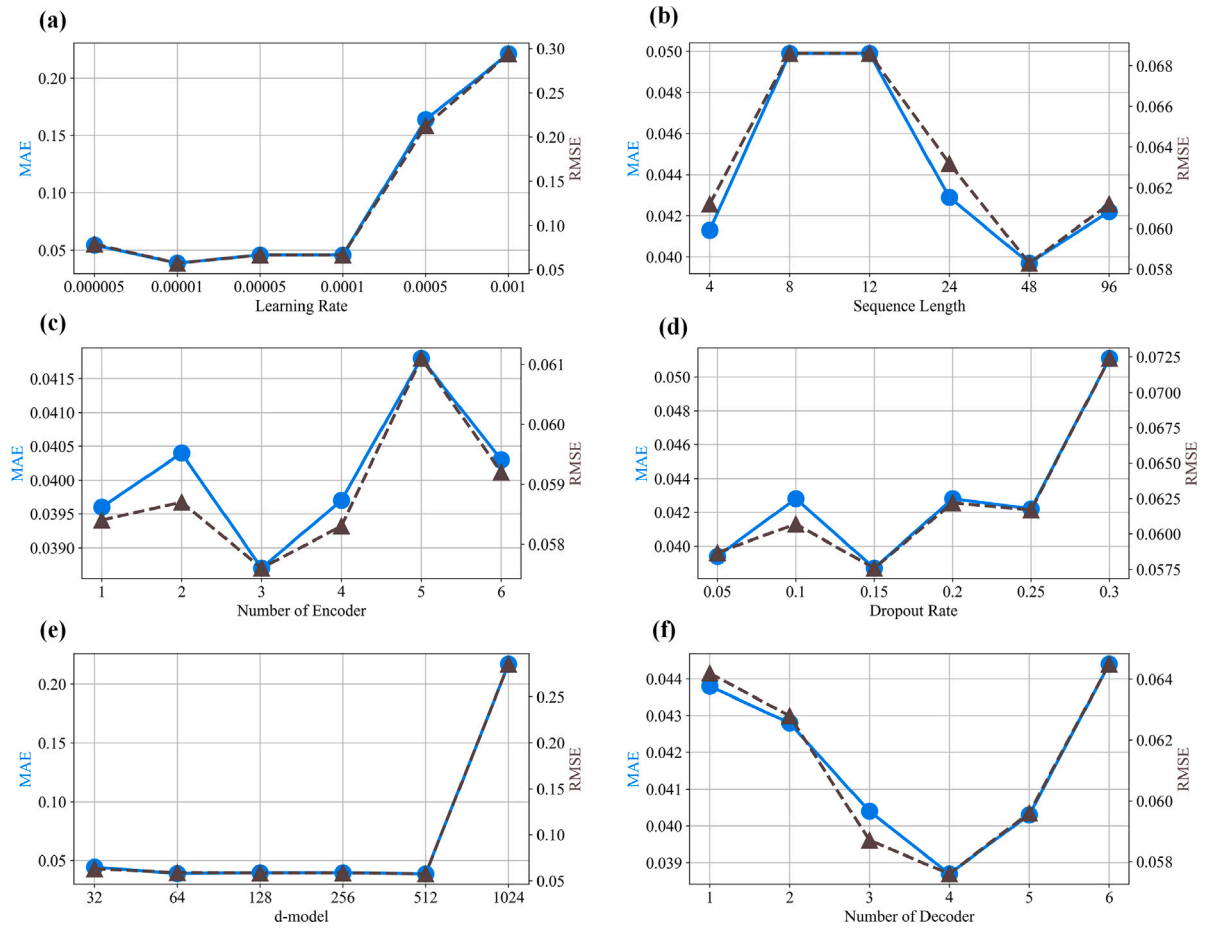


Fig. 10. The variations of hyper-parameter selection using the greedy algorithm.

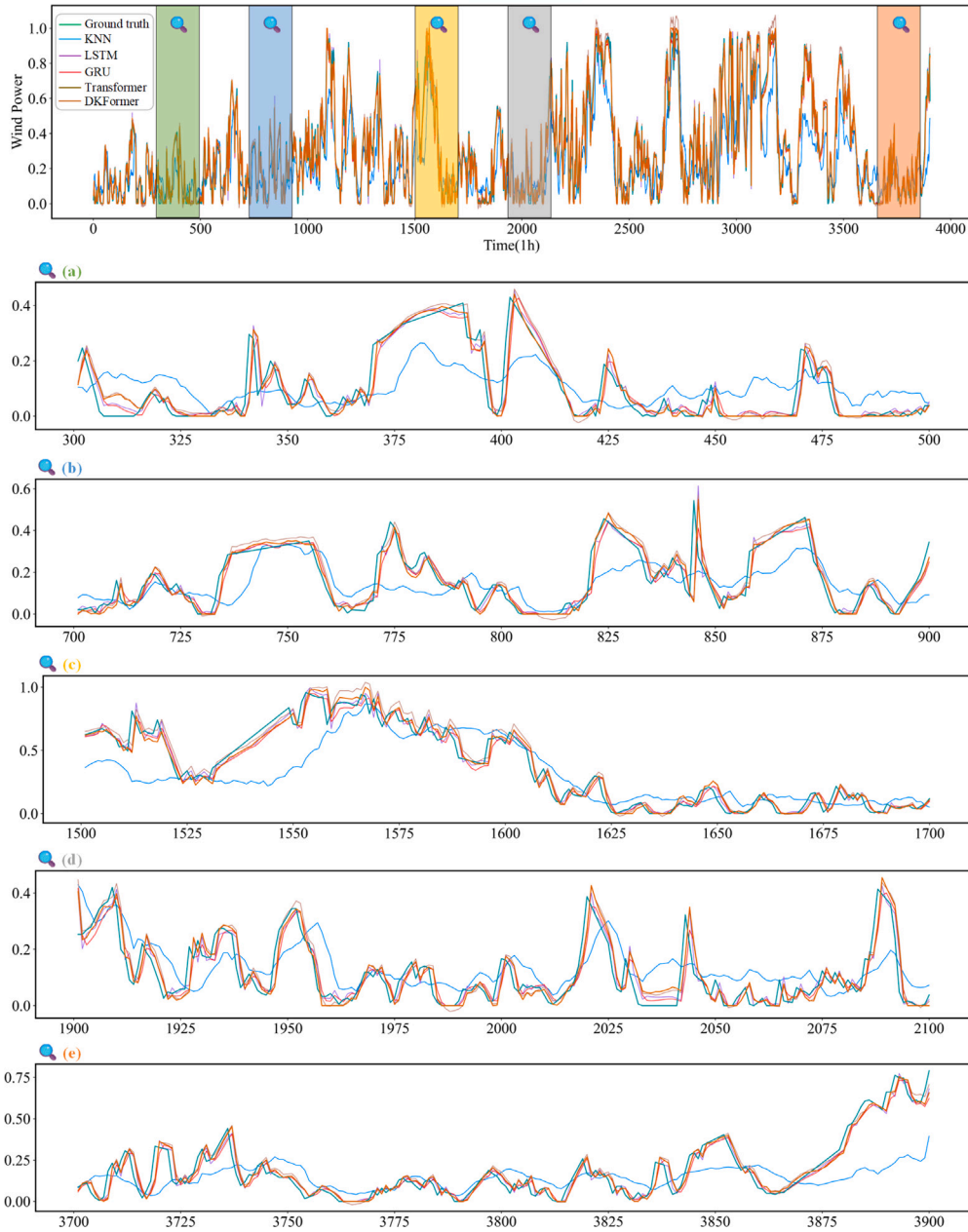


Fig. 11. The wind power forecasting comparisons on the target wind cluster between different models.

slightly increases the prediction time by 1 millisecond compared to the Transformer model, this marginal difference can be considered negligible in practical scenarios.

4.3. Comparative analysis

4.3.1. One-step forecasting

To evaluate the performance of DKFormer, a comparative analysis is presented in Fig. 11, showcasing its forecasting outcomes in comparison to conventional statistical and DL models such as KNN, LSTM, GRU, and Transformer specifically for the target wind cluster. The results clearly demonstrate that DKFormer yields predictive outcomes that closely align with the actual value curve, thereby indicating its exceptional proficiency in single-step wind power forecasting.

It is worth noting that wind power sequences often exhibit periods of continuous zero values, indicating the absence or low magnitude of wind speeds during certain time intervals. DKFormer possesses a

distinct advantage in fitting data with such characteristics. This advantage in one-step forecasting can be attributed to the incorporation of a clipping module in DKFormer, which is specifically designed to ensure adherence to physical laws when making predictions. The specially designed clipping module effectively prevents the generation of predictions that fall below 0 or exceed the maximum rated power, thereby enhancing both accuracy and reliability of the model.

4.3.2. Multi-step forecasting

Building upon previous studies [22,46], the multi-step wind power forecasting experiment encompasses both intra-day (less than 8 h) and short-term (8 to 48 h ahead) prediction horizons. This approach facilitates an extensive evaluation of the proposed DKFormer model's performance across various time scales. As shown in Table 2, the results clearly demonstrates that the DKFormer model outperforms other models in terms of accuracy and stability for all forecast horizons.

Table 2
Numerical results of multi-step wind power forecasting by different models.

Time step	Criteria($\times 10^{-1}$)	Forecasting models					
		KNN	GRU	LSTM	Transformer	DKFormer	TL+DKFormer
1 h	MAE (kW)	1.293	0.417 $\pm$ 0.018	0.391 $\pm$ 0.020	0.435 $\pm$ 0.032	0.387 $\pm$ 0.007	0.340 $\pm$ 0.001
	RMSE (kW)	1.706	0.610 $\pm$ 0.015	0.582 $\pm$ 0.026	0.627 $\pm$ 0.034	0.577 $\pm$ 0.011	0.524 $\pm$ 0.002
	R ²	6.079	9.499 $\pm$ 0.023	9.543 $\pm$ 0.047	9.470 $\pm$ 0.058	9.552 $\pm$ 0.018	9.621 $\pm$ 0.012
2 h	MAE (kW)	1.334	0.551 $\pm$ 0.028	0.509 $\pm$ 0.032	0.574 $\pm$ 0.057	0.505 $\pm$ 0.011	0.449 $\pm$ 0.003
	RMSE (kW)	1.759	0.788 $\pm$ 0.025	0.734 $\pm$ 0.029	0.832 $\pm$ 0.047	0.729 $\pm$ 0.021	0.637 $\pm$ 0.008
	R ²	5.834	9.163 $\pm$ 0.046	9.235 $\pm$ 0.061	9.067 $\pm$ 0.073	9.244 $\pm$ 0.044	9.426 $\pm$ 0.023
4 h	MAE (kW)	1.407	0.758 $\pm$ 0.055	0.724 $\pm$ 0.060	0.783 $\pm$ 0.087	0.681 $\pm$ 0.022	0.591 $\pm$ 0.009
	RMSE (kW)	1.854	1.074 $\pm$ 0.062	1.025 $\pm$ 0.065	1.099 $\pm$ 0.080	0.993 $\pm$ 0.045	0.821 $\pm$ 0.012
	R ²	5.370	8.446 $\pm$ 0.170	8.586 $\pm$ 0.183	8.372 $\pm$ 0.210	8.672 $\pm$ 0.129	9.124 $\pm$ 0.026
8 h	MAE (kW)	1.527	1.027 $\pm$ 0.108	0.985 $\pm$ 0.122	1.014 $\pm$ 0.197	0.900 $\pm$ 0.051	0.761 $\pm$ 0.014
	RMSE (kW)	2.012	1.407 $\pm$ 0.158	1.354 $\pm$ 0.152	1.242 $\pm$ 0.342	1.280 $\pm$ 0.099	1.076 $\pm$ 0.009
	R ²	4.549	7.334 $\pm$ 0.529	7.531 $\pm$ 0.580	7.582 $\pm$ 0.552	7.792 $\pm$ 0.385	8.563 $\pm$ 0.031
12 h	MAE (kW)	1.622	1.206 $\pm$ 0.176	1.177 $\pm$ 0.189	1.156 $\pm$ 0.226	1.028 $\pm$ 0.080	1.420 $\pm$ 0.011
	RMSE (kW)	2.134	1.609 $\pm$ 0.233	1.607 $\pm$ 0.254	1.564 $\pm$ 0.284	1.452 $\pm$ 0.150	1.423 $\pm$ 0.011
	R ²	3.870	6.515 $\pm$ 0.982	6.524 $\pm$ 1.162	6.703 $\pm$ 1.037	7.163 $\pm$ 0.682	7.341 $\pm$ 0.040
24 h	MAE (kW)	1.805	1.481 $\pm$ 0.299	1.556 $\pm$ 0.332	1.334 $\pm$ 0.296	1.206 $\pm$ 0.146	1.040 $\pm$ 0.036
	RMSE (kW)	2.368	1.965 $\pm$ 0.473	2.133 $\pm$ 0.504	1.788 $\pm$ 0.508	1.696 $\pm$ 0.261	1.462 $\pm$ 0.017
	R ²	2.458	4.803 $\pm$ 1.192	3.881 $\pm$ 1.217	5.693 $\pm$ 1.083	6.129 $\pm$ 0.884	6.981 $\pm$ 0.078
48 h	MAE (kW)	1.971	1.787 $\pm$ 0.403	1.749 $\pm$ 0.417	1.501 $\pm$ 0.397	1.381 $\pm$ 0.245	1.288 $\pm$ 0.039
	RMSE (kW)	2.600	2.370 $\pm$ 0.699	2.329 $\pm$ 0.721	2.193 $\pm$ 0.629	1.926 $\pm$ 0.405	1.724 $\pm$ 0.021
	R ²	0.900	2.439 $\pm$ 1.902	2.699 $\pm$ 1.817	3.982 $\pm$ 1.429	5.005 $\pm$ 1.212	5.462 $\pm$ 0.098

First of all, DKFormer outperforms other baseline models in terms of stability. The standard Transformer introduces self-attention and multi-head attention mechanisms, which involve a large number of learnable parameters and reduce the dependency on the distribution of training data. Building upon this, DKFormer further enhances the stability by utilizing additional domain knowledge constraints.

Moreover, the proposed method demonstrates high accuracy in multi-step wind power forecasting. It is worth noting that the forecasting performance of each model tends to decrease as the forecasting horizon increases. This decline can be attributed to the accumulation of errors in multi-step predictions. The multi-head attention mechanism employed in the Transformer model plays a crucial role in its superior performance, particularly in longer-term predictions. This mechanism enables parallel learning of sequential features at different time steps, thereby enhancing the model's ability to capture long-term dependencies. On the other hand, models based on recurrent neural networks (RNN), such as LSTM and GRU, exhibit better performance in scenarios with shorter time steps. This difference arises partly because RNN models process sequences step by step and lack the ability to parallelize the computation of the entire sequence.

In contrast, the statistical model KNN performs poorly in both single-step and multi-step wind power forecasting. Statistical models often struggle to capture complex relationships within sequences, as is evident from Fig. 11. KNN can only capture the general trend of the sequence and fails to effectively track real-time changes in wind power.

Additionally, the multi-step forecasting results are visualized in Fig. 12, showcasing the pronounced advantages in multi-step forecasting. It demonstrates remarkable responsiveness to abrupt changes in wind speed data while exhibiting a significantly reduced time delay effect compared to other baseline models. In contrast, other models merely capture the overall average trend of wind speed variations.

DKFormer, incorporating domain knowledge constraints on top of the Transformer architecture, surpasses both statistical models, and DL models based on RNN and Transformer. It not only implicitly captures short-term and long-term dependencies in wind power sequences but also explicitly integrates domain knowledge to guide the model's training and output. As a result, DKFormer achieves superior forecasting performance compared to other models in this study.

To further enhance the prediction accuracy and stability of the model, additional experiments on transfer learning are conducted. We incorporate transfer learning into the DKFormer model to construct the TL+DKFormer model. In this process, one wind cluster is randomly selected as the target data, while the data from the remaining six wind

clusters are utilized as the source data. The pre-training phase involves training DKFormer on the source data, which consists of a total of 52,704 data pairs. According to Table 2 results, compared to basic DKFormer without transfer learning, the constructed TL+DKFormer model demonstrates improved forecasting accuracy and stability.

4.4. Ablation experiment

The DKFormer model is a specialized architecture specifically designed for wind power forecasting, leveraging the Transformer framework and integrating three portable domain knowledge modules. Among them, the denoising module enhances forecasting accuracy by eliminating noise from the NWP data. The clipping module incorporates physical laws of wind power generation to enforce constraints during the model training process. The boundary module utilizes the lead time of NWP and the wind power generation curve to improve tracking performance in multi-step forecasting through boundary constraints.

In order to demonstrate the effectiveness and indispensability of each module, a series of ablation experiments are conducted. These experiments involve the removing individual modules from the system to observe their impact on overall performance, thereby revealing the roles and values of these components. In the ablation experiments, a baseline group and three experimental groups are considered. The baseline group includes the complete model with all components retained, while the three experimental groups remove the denoising, clipping, and boundary modules from the DKFormer, respectively (denoted as "w/o denoising", "w/o clipping" and "w/o boundary"). By comparing the performance differences between the baseline group and the experimental groups, the impact of each component on the model's performance can be determined.

The results of the ablation experiment are presented in Table 3 and Fig. 13. Eliminating any of the physical knowledge constraint modules from the DKFormer leads to a deterioration in all three performance metrics of the model, indicating that each module plays a distinct role by effectively incorporating domain knowledge constraints into the data pre-processing, model training, and forecasting output stages. Consequently, the decisions made by the DL model become more rationalized. The boundary module is primarily designed to address error accumulation in multi-step forecasting, thus its effectiveness becomes more pronounced with an increase in forecasting stride.

In addition, we further illustrate the outcomes of ablation experiments in Fig. 14. The figure clearly demonstrates that the phenomenon of time delay is exacerbated when any module is eliminated from

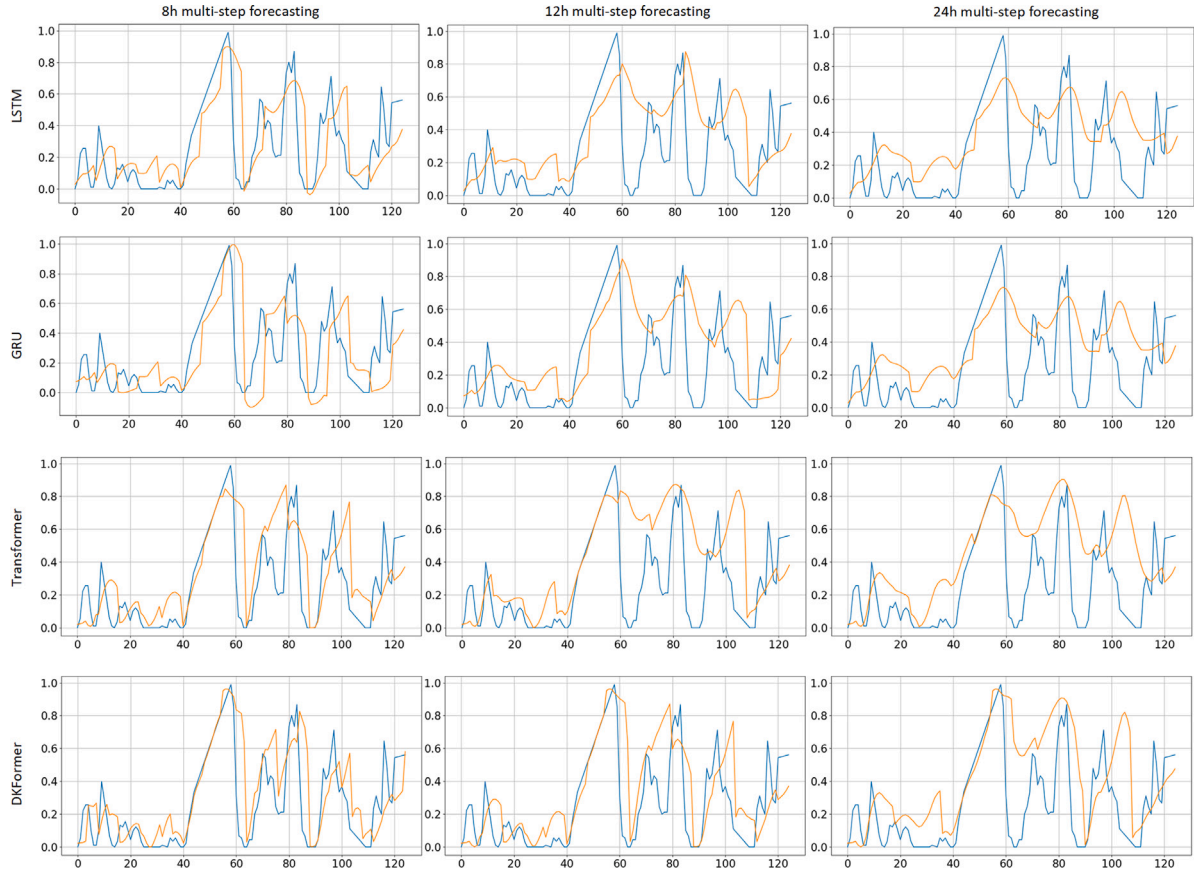


Fig. 12. Visualization of multi-step wind power forecasting by different models.

Table 3
Numerical results of ablation experiments.

Time step	Criteria($\times 10^{-1}$)	Forecasting models			
		w/o denoising	w/o clipping	w/o boundary	DKFormer
1 h	MAE (kW)	0.392 ± 0.013	0.395 ± 0.014	0.389 ± 0.007	0.387 ± 0.007
	RMSE (kW)	0.582 ± 0.015	0.586 ± 0.017	0.579 ± 0.011	0.577 ± 0.011
	R^2	9.530 ± 0.023	9.490 ± 0.026	9.552 ± 0.026	9.552 ± 0.018
24 h	MAE (kW)	1.283 ± 0.183	1.304 ± 0.207	1.267 ± 0.166	1.206 ± 0.146
	RMSE (kW)	1.715 ± 0.384	1.730 ± 0.392	1.702 ± 0.328	1.696 ± 0.261
	R^2	4.594 ± 0.895	4.134 ± 0.892	6.060 ± 0.889	6.129 ± 0.884

DKFormer. This can be attributed to the fact that each module in DKFormer fulfills a distinct role, and removing a module disrupts the seamless functioning of the model, leading to either malfunction or adverse effects on other modules.

4.5. Module compatible evaluation

In order to evaluate the compatibility of the proposed portable modules with other DL models, a series of module compatibility assessments are conducted. The denoising, clipping, and boundary modules are individually integrated into LSTM and GRU models to assess their effectiveness. Following the parameter settings in Section 4.2, the capabilities of these models on both the one-step and day-ahead forecasting scales are evaluated. The results presented in Table 4 clearly demonstrate that incorporating any of these modules leads to significant performance improvements. Particularly, the denoising modules exhibit remarkable efficacy by substantially enhancing performance in both (1-step) and longer-term (day-ahead) forecasting tasks. On the other hand, the boundary module primarily functions effectively when dealing with extended time steps due to potential time delay issues

leading to predictions exceeding boundaries. The day-ahead forecasting outcomes suggest that incorporating the boundary module effectively mitigates such time delays.

5. Conclusion

Wind power generation forecasting is crucial for making deployment decisions of wind power plants, especially when faced with limited data. In this paper, we proposed DKFormer, a transformer-based forecasting model that integrates DL with domain knowledge through three portable modules: denoising, clipping, and boundary. These modules played essential roles in data pre-processing, model training, and forecasting stages, respectively. Transfer learning methods were employed to address the issue of limited training data and enhance the accuracy and stability of the forecasts. Furthermore, DKFormer utilized the cubic boundary curves to leverage the future NWP data to reduce time delay and increase prediction accuracy.

The experiments conducted on real-life datasets have demonstrated that DKFormer outperforms conventional statistical and DL models in both intra-day and short-term forecasting, particularly for cases with

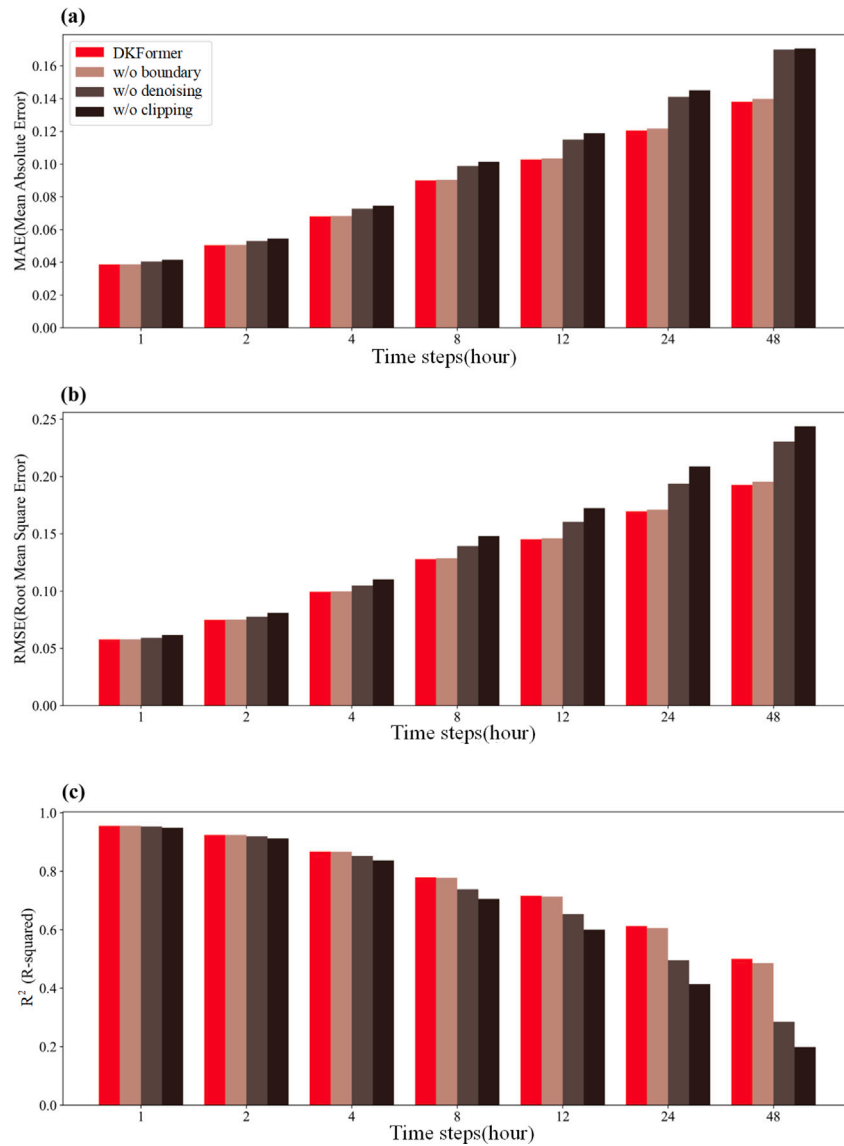


Fig. 13. The results of ablation experiments.

Table 4

Numerical results of multi-step wind power forecasting by different models with portable module.

Time step	Criteria($\times 10^{-1}$)	Forecasting models							
		LSTM	LSTM + denoising	LSTM + clipping	LSTM + boundary	GRU	GRU+denoising	GRU+clipping	GRU+boundary
1 h	MAE (kW)	0.391 $\pm$ 0.020	0.362 $\pm$ 0.016	0.368 $\pm$ 0.017	0.391 $\pm$ 0.020	0.417 $\pm$ 0.018	0.392 $\pm$ 0.016	0.412 $\pm$ 0.017	0.417 $\pm$ 0.018
	RMSE (kW)	0.582 $\pm$ 0.026	0.546 $\pm$ 0.025	0.552 $\pm$ 0.024	0.582 $\pm$ 0.026	0.610 $\pm$ 0.015	0.573 $\pm$ 0.014	0.605 $\pm$ 0.013	0.610 $\pm$ 0.015
	R^2	9.543 $\pm$ 0.047	9.620 $\pm$ 0.045	9.589 $\pm$ 0.047	9.543 $\pm$ 0.047	9.499 $\pm$ 0.023	9.582 $\pm$ 0.022	9.509 $\pm$ 0.021	9.499 $\pm$ 0.023
24 h	MAE (kW)	1.556 $\pm$ 0.332	1.437 $\pm$ 0.323	1.486 $\pm$ 0.329	1.502 $\pm$ 0.332	1.481 $\pm$ 0.299	1.379 $\pm$ 0.280	1.399 $\pm$ 0.290	1.471 $\pm$ 0.295
	RMSE (kW)	2.133 $\pm$ 0.504	2.002 $\pm$ 0.487	2.009 $\pm$ 0.499	2.112 $\pm$ 0.500	1.965 $\pm$ 0.473	1.769 $\pm$ 0.469	1.778 $\pm$ 0.468	1.867 $\pm$ 0.470
	R^2	3.881 $\pm$ 1.217	4.378 $\pm$ 1.202	4.299 $\pm$ 1.206	4.221 $\pm$ 1.210	4.803 $\pm$ 1.192	5.041 $\pm$ 1.186	5.022 $\pm$ 1.188	4.988 $\pm$ 1.190

larger time steps. We have observed significant improvements in MAE compared to the vanilla Transformer model, with reductions of 22.0% for 24-h forecasting. Furthermore, our ablation experiments highlight the indispensability and effectiveness of the three domain knowledge constraint modules within DKFormer. Each module played a unique and vital role in incorporating domain knowledge, thereby further enhancing the forecasting performance of the model.

In the future, the integration of the proposed three domain knowledge injection modules into other base models holds great promise

due to their high portability and potential for further research exploration. In addition, our paper introduced a pioneering approach in wind power generation models by concurrently leveraging historical wind power data and future NWP wind speed data for forecasting purposes. However, considering the limited forecast horizon of NWP data (typically 48 h), future research could investigate concurrent prediction of both wind power and wind speed to enable long-term forecasting applications, thus advancing the field.

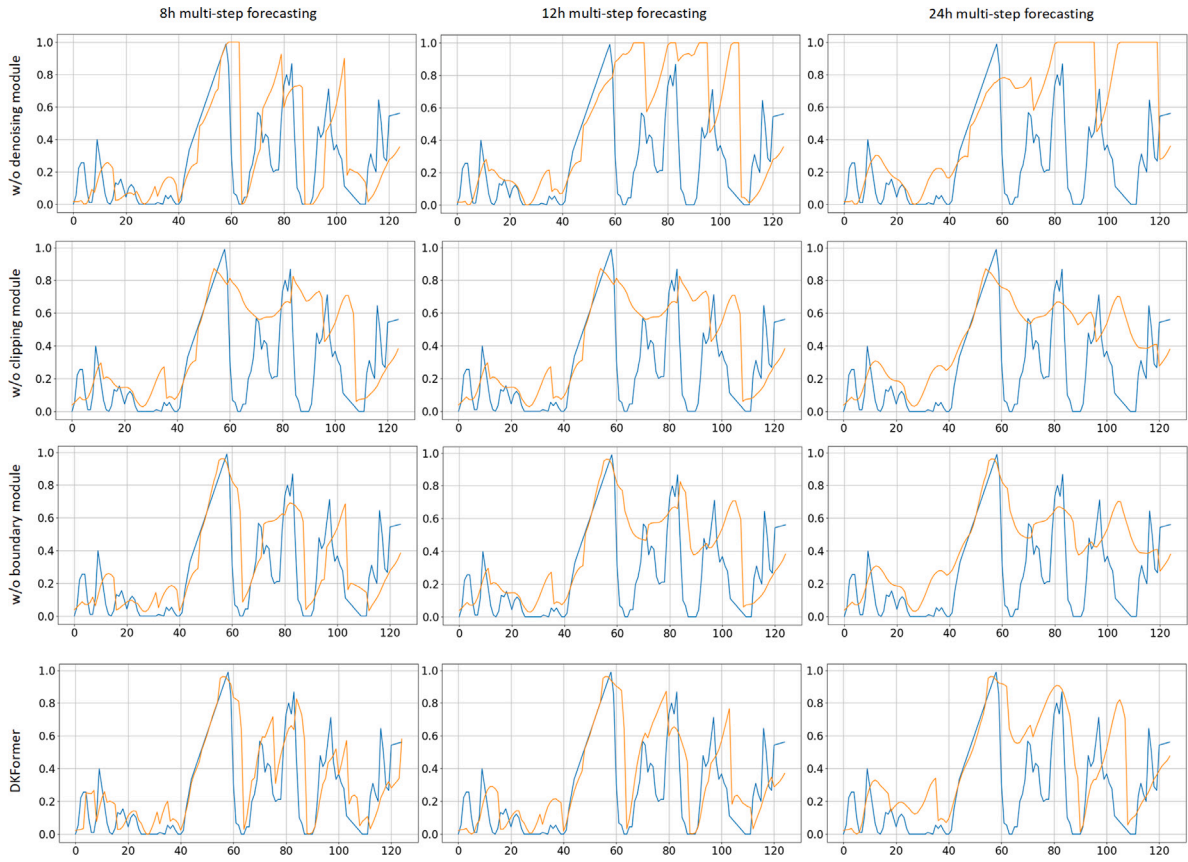


Fig. 14. The visualization results of ablation experiments.

CRedit authorship contribution statement

Junhao Cheng: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Xing Luo:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Zhi Jin:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported and partially funded by: the Major Key Project of Peng Cheng Laboratory (Grant No. PCL2023A09); the National Natural Science Foundation of China (Grant No. 62071500); Shenzhen Science and Technology Program (Grant No. JCYJ2023080 7111107015); and the Fundamental Research Funds for the Central Universities, Sun Yat-sen University (Grant No. 41gqb015).

Appendix

According to the analysis conducted in Section 3.1.3, the upper and lower boundary functions in the boundary module can be expressed as:

$$f(x) = p_1 * x^3 + p_2 * x^2 + p_3 * x + p_4 + p_5 * e^{(p_6 * x)} \quad (\text{A.1})$$

To pre-process the data, the widely-used DBSCAN clustering algorithm is employed for outlier removal [47,48]. The DBSCAN algorithm relies on two key parameters: ϵ , which defines the radius of a

point's neighborhood, and $minPts$, which indicates the minimum number of points required for cluster formation. Consistent with previous studies [47,48], we set ϵ as 3 and $minPts$ as 2 in this study.

Afterwards, the upper and lower boundary functions, $f_{lower}(x)$ and $f_{upper}(x)$, can be easily fitted onto the target dataset using the “cftool” toolbox in MATLAB. According to Eq. (11), it is worth noticing that p_1 and p_3 should be positive after polynomial expansion. Therefore, a minimum value of 0 is set for both variables p_1 and p_3 during the auto-fitting process. Additionally, an exponential term has been incorporated as a correction factor to mitigate the impact of forecasting errors. The upper and lower boundary functions can be determined through the following steps.

1. Run the DBSCAN algorithm in Matlab with parameters $\epsilon = 3$ and $minPts = 2$. The algorithm will classify each point as either belonging to a cluster or being an outlier.
2. Based on the labels generated by the algorithm, eliminate all points identified as outliers (typically labeled -1).
3. Obtain the NWP data with outliers removed.
4. Import the remaining NWP data points into “cftool” toolbox, select cubic polynomial fitting, and specify a constraint that p_1 and p_3 should not be less than 0.

Consequently, the final upper and lower boundary functions for the target wind cluster are presented in Table A.1, while the fitting result is depicted in Fig. 5.

Data availability

Data will be made available on request.

Table A.1
The determined parameters of upper and lower boundary functions for the target wind cluster.

Parameter	p_1	p_2	p_3	p_4	p_5	p_6
$f_{lower}(x)$	2.606	-1.035	0.001	-0.001	0.011	-3
$f_{upper}(x)$	1.373	-2.314	0.870	1.047	-1.043	-6

References

[1] Bp B. Statistical review of world energy 2022. 2023.

[2] Zhang Y, Liu K, Qin L, An X. Deterministic and probabilistic interval prediction for short-term wind power generation based on variational mode decomposition and machine learning methods. *Energy Convers Manage* 2016;112:208–19.

[3] Xu Y, Jia L, Yang W. Correlation based neuro-fuzzy Wiener type wind power forecasting model by using special separate signals. *Energy Convers Manage* 2022;253:115173.

[4] Shen Y, Wang X, Chen J. Wind power forecasting using multi-objective evolutionary algorithms for wavelet neural network-optimized prediction intervals. *Appl Sci* 2018;8(2):185.

[5] Wang Y, Hu Q, Li L, Foley AM, Srinivasan D. Approaches to wind power curve modeling: A review and discussion. *Renew Sustain Energy Rev* 2019;116:109422.

[6] Yan J, Ouyang T. Advanced wind power prediction based on data-driven error correction. *Energy Convers Manage* 2019;180:302–11.

[7] Jiang Y, Song Z, Kusiak A. Very short-term wind speed forecasting with Bayesian structural break model. *Renew Energy* 2013;50:637–47.

[8] Song Z, Jiang Y, Zhang Z. Short-term wind speed forecasting with Markov-switching model. *Appl Energy* 2014;130:103–12.

[9] Yang L, He M, Zhang J, Vittal V. Support-vector-machine-enhanced markov model for short-term wind power forecast. *IEEE Trans Sustain Energy* 2015;6(3):791–9.

[10] Khosravi A, Nahavandi S. Combined nonparametric prediction intervals for wind power generation. *IEEE Trans Sustain Energy* 2013;4(4):849–56.

[11] Qian Y, Wang T, Yuan Y, Zhang Y. Comparative study on wind turbine wakes using a modified partially-averaged Navier-Stokes method and large eddy simulation. *Energy* 2020;206:118147.

[12] Zhou H, Qiu Y, Feng Y, Liu J. Power prediction of wind turbine in the wake using hybrid physical process and machine learning models. *Renew Energy* 2022;198:568–86.

[13] Aggarwal SK, Gupta M. Wind power forecasting: A review of statistical models-wind power forecasting. *Int J Energy Sci* 2013;3(1):1–10.

[14] Wang Y, Zou R, Liu F, Zhang L, Liu Q. A review of wind speed and wind power forecasting with deep neural networks. *Appl Energy* 2021;304:117766.

[15] Erdem E, Shi J. ARMA based approaches for forecasting the tuple of wind speed and direction. *Appl Energy* 2011;88(4):1405–14.

[16] Liu H, Tian H-q, Li Y-f. Comparison of two new ARIMA-ANN and ARIMA-Kalman hybrid methods for wind speed prediction. *Appl Energy* 2012;98:415–24.

[17] Hu J, Wang J, Ma K. A hybrid technique for short-term wind speed prediction. *Energy* 2015;81:563–74.

[18] Mahaseth R, Kumar N, Aggarwal A, Tayal A, Kumar A, Gupta R. Short term wind power forecasting using k-nearest neighbour (KNN). *J Inf Optim Sci* 2022;43(1):251–9.

[19] Louka P, Galanis G, Siebert N, Kariniotakis G, Katsafados P, Pytharoulis I, et al. Improvements in wind speed forecasts for wind power prediction purposes using Kalman filtering. *J Wind Eng Ind Aerodyn* 2008;96(12):2348–62.

[20] Chen N, Qian Z, Nabney IT, Meng X. Wind power forecasts using Gaussian processes and numerical weather prediction. *IEEE Trans Power Syst* 2013;29(2):656–65.

[21] Kariniotakis G, Nogaret E, Stavrakakis G. A fuzzy logic and a neural network based wind power forecasting model. *Wind Energy* 1996;20:24.

[22] Huang Y, Liu G-P, Hu W. Priori-guided and data-driven hybrid model for wind power forecasting. *ISA Trans* 2023;134:380–95.

[23] Girshick R, Donahue J, Darrell T, Malik J. Rich feature hierarchies for accurate object detection and semantic segmentation. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2014, p. 580–7.

[24] Shahid F, Zameer A, Muneeb M. A novel genetic LSTM model for wind power forecast. *Energy* 2021;223:120069.

[25] Ko M-S, Lee K, Kim J-K, Hong CW, Dong ZY, Hur K. Deep concatenated residual network with bidirectional LSTM for one-hour-ahead wind power forecasting. *IEEE Trans Sustain Energy* 2020;12(2):1321–35.

[26] Niu Z, Yu Z, Tang W, Wu Q, Reformat M. Wind power forecasting using attention-based gated recurrent unit network. *Energy* 2020;196:117081.

[27] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need. *Adv Neural Inf Process Syst* 2017;30.

[28] Qu K, Si G, Shan Z, Kong X, Yang X. Short-term forecasting for multiple wind farms based on transformer model. *Energy Rep* 2022;8:483–90.

[29] Wang L, He Y, Liu X, Li L, Shao K. M2TNet: Multi-modal multi-task transformer network for ultra-short-term wind power multi-step forecasting. *Energy Rep* 2022;8:7628–42.

[30] Sun S, Liu Y, Li Q, Wang T, Chu F. Short-term multi-step wind power forecasting based on spatio-temporal correlations and transformer neural networks. *Energy Convers Manage* 2023;283:116916.

[31] Luo X, Zhang D, Zhu X. Deep learning based forecasting of photovoltaic power generation by incorporating domain knowledge. *Energy* 2021;225:120240.

[32] Luo X, Zhang D. An adaptive deep learning framework for day-ahead forecasting of photovoltaic power generation. *Sustain Energy Technol Assess* 2022;52:102326.

[33] Luo X, Zhang D, Zhu X. Combining transfer learning and constrained long short-term memory for power generation forecasting of newly-constructed photovoltaic plants. *Renew Energy* 2022;185:1062–77.

[34] Luo X, Zhang D. A cascaded deep learning framework for photovoltaic power forecasting with multi-fidelity inputs. *Energy* 2023;268:126636.

[35] Wang J, Zhang W, Li Y, Wang J, Dang Z. Forecasting wind speed using empirical mode decomposition and Elman neural network. *Appl Soft Comput* 2014;23:452–9.

[36] Liu H, Chen C. Data processing strategies in wind energy forecasting models and applications: A comprehensive review. *Appl Energy* 2019;249:392–408.

[37] Abedinia O, Lotfi M, Bagheri M, Sobhani B, Shafie-Khah M, Catalão JP. Improved EMD-based complex prediction model for wind power forecasting. *IEEE Trans Sustain Energy* 2020;11(4):2790–802.

[38] Li F, Liao H. An intelligent method for wind power forecasting based on integrated power slope events prediction and wind speed forecasting. *IEEE Trans Electr Electron Eng* 2018;13(8):1099–105.

[39] Gao J, Chen Y, Hu W, Zhang D. An adaptive deep-learning load forecasting framework by integrating transformer and domain knowledge. *Adv Appl Energy* 2023;10:100142.

[40] Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T. Transformers for image recognition at scale. 2020, arXiv preprint arXiv:2010.11929.

[41] Wen Q, Zhou T, Zhang C, Chen W, Ma Z, Yan J, et al. Transformers in time series: A survey. 2022, arXiv preprint arXiv:2202.07125.

[42] Wu H, Meng K, Fan D, Zhang Z, Liu Q. Multistep short-term wind speed forecasting using transformer. *Energy* 2022;261:125231.

[43] Hansen M. *Aerodynamics of wind turbines*. Routledge; 2015.

[44] Ding M, Zhou H, Xie H, Wu M, Nakanishi Y, Yokoyama R. A gated recurrent unit neural networks based wind speed error correction model for short-term wind power forecasting. *Neurocomputing* 2019;365:54–61.

[45] Hong T, Pinson P, Fan S. Global energy forecasting competition 2012. *Int J Forecast* 2014;30(2):357–63.

[46] Tuncar EA, Sağlam Ş, Oral B. A review of short-term wind power generation forecasting methods in recent technological trends. *Energy Rep* 2024;12:197–209.

[47] Hou G, Wang J, Fan Y. Wind power forecasting method of large-scale wind turbine clusters based on DBSCAN clustering and an enhanced hunter-prey optimization algorithm. *Energy Convers Manage* 2024;307:118341.

[48] Wang S, Chen C. Short-term wind power prediction based on DBSCAN clustering and support vector machine regression. In: *2020 5th international conference on computer and communication systems*. IEEE; 2020, p. 941–5.